

Product Requirements Document

Kahoot — Real-Time Quiz & Learning Game Platform

Version: 1.1

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Author: [Author]

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1. Overview

1.1 Product Summary

A real-time, multiplayer quiz platform that lets hosts create and run interactive quiz games for live audiences. Players join via a game PIN on any device, answer timed questions, and compete on a live leaderboard — no account required to play.

1.2 Problem Statement

Traditional knowledge checks (polls, quizzes, tests) are passive and disengaging. Instructors, trainers, and facilitators need a low-friction, high-energy way to assess understanding and keep audiences active during sessions.

1.3 Goals

- Enable hosts to create a quiz in under 5 minutes
- Support up to 150 concurrent players per game
- Deliver a seamless, real-time experience across mobile and desktop
- Keep the barrier to play near zero (no app download, no login)

1.4 Out of Scope (v1)

- Team/group mode
- Audio or video in questions
- Analytics dashboard beyond session-level stats

- Offline mode
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2. Users & Roles

Role	Description
Host	Creates quizzes, launches game sessions, controls pacing
Player	Joins via PIN, answers questions in real time
Guest Host	Can run an existing quiz without an account (stretch)

2.1 Host Persona

Teachers, corporate trainers, workshop facilitators, and event emcees. Needs to prepare content ahead of time and run a game live with minimal friction. Values reliability and control over the game flow.

2.2 Player Persona

Students, workshop attendees, event participants. Joins on a phone or laptop. Expects an instant, familiar, low-stakes experience. No tolerance for login friction.

3. Core Features

3.1 Quiz Builder

Hosts can create and manage quizzes from a dashboard. Anonymous (unsaved) quizzes are also supported for one-off use without an account.

Requirements:

- Create a quiz with a title, optional description, and cover image
- **Anonymous quiz mode:** Host can build and run a quiz in a single session without saving it or logging in. Quiz is discarded after the session ends.
- Add, reorder, and delete questions
- Question types supported in v1:
 - Multiple choice (2–4 options, single correct answer)

- True/False
- Per-question settings:
 - Question text (required, max 120 chars)
 - Answer options (required)
 - Correct answer designation (required)
 - Time limit: 5, 10, 20, 30, 60, 90, 120 seconds (default: 20s)
 - Point value: standard (1000 pts) or double points
- Media: optional image per question (upload or URL)
- Autosave while editing
- Duplicate and delete quizzes

3.2 Game Session — Host View

The host starts a game from any quiz and manages the session from a dedicated host screen.

Lobby phase:

- System generates a unique 6-digit game PIN
- Host screen shows PIN prominently with a player join list that updates in real time
- Host can kick players before the game starts
- Host manually advances from lobby to start

In-game phase:

- Host controls question advancement (manual “next” or auto-advance option)
- Host screen shows:
 - Current question text and options (dimmed — not shown to players until revealed)
 - Countdown timer
 - Live response count (X of Y players answered)
 - Answer distribution bar chart after time expires
 - Correct answer reveal with player count per option

- After each question: intermediate podium showing top 3 players

End of game:

- Final leaderboard (all players ranked)
- Option to export results as CSV
- Option to replay the same quiz

3.3 Presenter Mode (Projection View)

When a host launches a session, two URLs are generated:

- **Control panel** (`/host/[gamePIN]/control`) — the host's private screen. Shows player list, answer distribution, correct answer reveal, and game controls (next question, end game). Optimized for a laptop.
- **Projection view** (`/host/[gamePin]/display`) — a minimal, audience-facing screen. Designed to be opened in a second window and shared via display mirroring or a projector. Shows:
 - Game PIN and join URL prominently during lobby
 - Question text (large, readable from the back of a room)
 - Countdown timer
 - Answer option shapes/colors (no text — players see text on their own devices)
 - Answer distribution bars after time expires
 - Leaderboard podium between questions

The host opens both URLs at session start. The control panel stays on the laptop; the display view goes fullscreen on the projected screen. No plugins, no embeds required.

3.4 Game Session — Player View

Players join at `kahoot.app/join` (or equivalent), enter the PIN, and pick a nickname.

Lobby: Waiting screen with nickname displayed, animated until game starts.

In-game:

- Question text displayed for a configurable reveal window (e.g., 3s) before options appear
- Answer options shown as large colored buttons (matching Kahoot's iconic 4-color grid):

red triangle, blue circle, yellow star, green square

- Timer countdown visible
- After answering: locked-in confirmation screen (cannot change answer)
- After time expires: feedback screen showing correct/incorrect + points earned this round + current rank

End of game:

- Final rank and total score
- Option to share score (stretch)

3.5 Leaderboard & Scoring

- Players earn points based on: correctness + speed
- **Formula:** $\text{points} = \text{base_value} \times (\text{time_remaining} / \text{time_limit}) \times \text{speed_multiplier}$
- Streak bonuses for consecutive correct answers (stretch goal)
- Leaderboard updates after every question
- Top 3 players shown on podium between questions (host screen only)

3.6 Authentication (Host Only)

- Email/password signup and login
- OAuth: Google sign-in
- Password reset via email
- Players never need an account

3.7 Pricing & Player Limits

Free tier is unlimited in games and quizzes, gated on players per session.

Tier	Player Cap	Price
Free	10 players/session	\$0
Pro	150 players/session	TBD

- The 150-player cap is a hard system limit across all tiers

- When a free session reaches 10 players, new join attempts are rejected with a friendly upsell prompt
 - Anonymous quiz sessions follow the same player cap rules as free accounts
 - Upsell prompt appears in the host lobby when the player count approaches the free limit
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4. Technical Requirements

4.1 Real-Time Communication

- WebSocket-based connection for all in-game state (question delivery, answer submission, leaderboard updates)
- Reconnection handling: players who drop can rejoin mid-game by re-entering PIN + nickname within a grace window
- Host disconnect: game pauses, auto-resumes when host reconnects within 60 seconds

4.2 Scalability

- Target: 150 concurrent players per room (hard cap), 1,000 concurrent rooms in v1
- Game state managed server-side; clients are thin
- Redis pub/sub or similar for room state management

4.3 Latency

- Answer submission acknowledged within 500ms under normal network conditions
- Leaderboard update delivered to all clients within 1s of question close

4.4 Platforms

- Web-first (responsive)
- Minimum browser support: Chrome 110+, Safari 16+, Firefox 110+
- No native app in v1

4.5 Data Persistence

- Quizzes: persisted per user account
- Game sessions: results stored for 30 days

- Player data: nicknames only, no PII collected from players
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5. UX & Design Principles

- **Immediacy:** Every action should feel instant. No loading screens during gameplay.
 - **Clarity:** Players should never be confused about what to do next.
 - **Energy:** Animations, color, and sound (stretch) reinforce the game feel.
 - **Accessibility:** Minimum WCAG 2.1 AA. Color options should not be the sole differentiator between answer choices — shapes/icons must also distinguish options.
 - **Mobile-first for players:** Player UI designed for one-handed use on phones.
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6. User Stories

Host

- As a host, I can create a new quiz from scratch so I can run it in my next session.
- As a host, I can launch a game from any saved quiz and get a shareable PIN.
- As a host, I can see how many players answered each option so I know where confusion lies.
- As a host, I can end the game early if needed.
- As a host, I can download session results as a CSV.

Player

- As a player, I can join a game with a PIN and a nickname, no account needed.
 - As a player, I can see the question and options clearly on my phone.
 - As a player, I get instant feedback on whether I was right and how I rank.
 - As a player, I can see my final score at the end of the game.
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7. Success Metrics

Metric	Target (3 months post-launch)
Quizzes created	1,000+
Game sessions run	500+
Avg players per session	15+
Session completion rate	>80%
Player join-to-answer rate	>90%
Host NPS	>40

8. Milestones

Milestone	Scope	Target
M1 — Core Loop	Quiz builder (saved + anonymous), single session, ≤10 players, presenter mode	Week 4
M2 — Scale & Polish	150 players/room, reconnect, CSV export, player cap enforcement	Week 8
M3 — Public Beta	Auth, Pro tier, quiz library, session history	Week 12

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
WebSocket reliability on mobile networks	Medium	High	Fallback polling; reconnect with grace period
Leaderboard scoring disputes (latency unfairness)	Low	Medium	Server-side timestamps for answer submission
Nickname abuse / inappropriate content	Medium	Medium	Client-side profanity filter + host kick capability
Cheating (answer interception)	Low	Low	Correct answers never

			sent to client until reveal
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10. Decisions Log

#	Question	Decision
1	Should hosts be able to set a max player limit per session?	No host-configurable limit. Hard system cap of 150 players/session across all tiers.
2	Do we support anonymous (unsaved) quizzes for one-off use?	Yes. Anonymous quizzes can be built and run without an account; discarded after the session ends.
3	Free tier model — unlimited games or gate on session size?	Gate on players. Free tier: 10 players/session. Pro tier: up to 150 players/session.
4	Host screen mirroring / Keynote integration?	Presenter mode: two separate URLs per session — a control panel (host’s laptop) and a projection view (fullscreen on projected display). No plugins required.

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